

Plug-and-Play Wearables: A Repositionable E-Textile Garment System to Support Custom Fit for Lower-Limb Rehabilitation Applications

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ABSTRACT

Fit of an e-textile garment has become increasingly important as more advanced wearable technology applications demand precise placement of sensors and actuators. Human anthropometry is complex and varied, and for many applications, a single e-textile garment cannot accurately fit a variety of users. One such example is lower-limb rehabilitation applications, which rely on precise placement of sensors and actuators with respect to joints and muscles for each individual user. Here, we present the development of a multi-layer, stretchable, flexible e-textile system, which affords quick and easy repositioning of components on the garment surface. Beyond custom fitting, this infrastructure also affords dynamic functionality of the garment, by adding and replacing sensors and actuators to enable a wide variety of applications.

CCS CONCEPTS

• **Hardware** → Emerging technologies; Circuit substrates; Flexible and printable circuits; Emerging technologies; Emerging interfaces; Communication hardware, interfaces and storage; Sensor applications and deployments.

KEYWORDS

E-textiles, wearable technology, wearable sensing, wearable actuation, lower-limb rehabilitation

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1 INTRODUCTION

As wearable technology applications evolve beyond single-point systems like wristbands, increasingly they demand specific positioning of sensing and actuating components on the body surface. Such applications often benefit from garment-integration, as the garment can serve to help position components so that setup is more efficient and enable shared power and networking. However, the garment structure itself often inhibits accuracy of device placement. Sizing and fit of e-textile garments is a complex problem, the implications of which are only beginning to be understood by the research and technology development communities [1], [2]. Affording the ability to customize the position of components within an e-textile garment structure would allow the benefits of garment-integration to be preserved, while simultaneously allowing for precise device placement for every user. Further, the ability to “plug and play” components on an e-textile garment also affords the ability to more quickly customize garment-integrated hardware systems, so that a broader variety of functionality can be quickly prototyped or deployed.

Repositionability, modularity, and customizability for e-textile systems are concepts that have been explored in a few prior projects, most commonly by providing multiple fixed attachment points for components [3]. The Tobiko system allows rigid PCB components to be placed anywhere on an e-textile pad and maintain power supplied through an array of diode-controlled contact pins (but does not provide a mechanical connection to the textile pad) [3]. Toolkit approaches frequently focus on “modularity” in the sense that PCB modules are developed to be stitchable or otherwise integrated into textiles, so that circuits can be built more quickly (but do not necessarily support repositionability) [4], [5]. Other modular approaches seek to facilitate more direct adjustment of the sizes of e-textile garments, by fabricating textile-integrated electronic panels that can be cut to size as the garment is fabricated [6], or fabricate textile-based electronics in strip form for insertion into pockets or adhered to a garment surface [7]. Finally, some approaches consider the attachment of components to textile-integrated bus architectures, but in a permanent manner that does not facilitate repositioning [8].

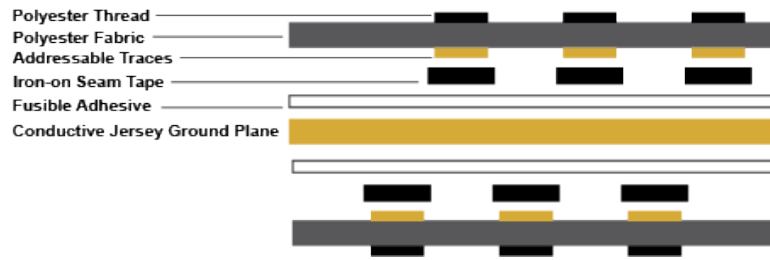


Figure 1: Cross-section of e-textile architecture

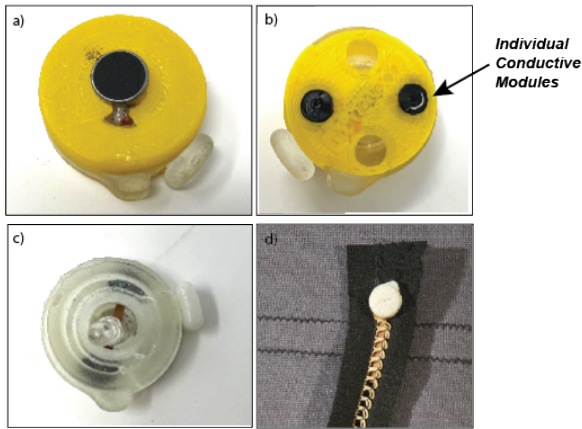


Figure 2: (a) and (b) LumeoLoop two-pin connector (with vibrating motor attached); (c) and (d) single-pin connector (connecting a textile strain sensor)

2 SYSTEM DESIGN

The garment system developed here relies on a multi-layer e-textile architecture, illustrated in Figure 1, and a detachable connector, illustrated in Figure 2 and Figure 3. This architecture uses textile materials to create addressable traces on the inside and outside of the garment, with a shared ground plane in the middle of the layup. Traces are insulated with non-conductive fabric and thread on the outside, and with tightly knitted seam tape on the inside.

The repositionable connector is adapted based on LumeoLoop Conductive attachment interface (LumeoTech), which attaches to the textile using conductive barbs that are retracted when the connector is detached. The textile layering provides a stand-off and mechanical barrier that prevents the connector from penetrating through both the addressable traces and the ground plane (which would create a short circuit), or through to the skin. We use three versions of the LumeoLoop connector (Figure 3): a non-conductive module, which allows textile components to be secured along the garment surface without making electrical connections (as shown in Figure 7); a single-pin module, which connects a single component lead (conductive thread, wire, soldering tab) to a conductive trace (either an addressable trace or the ground plane); and a two-pin module, which houses two conductive and two non-conductive LumeoLoop modules in one custom housing 3D printed with PLA filaments, and allows two pins of a component to be connected

to the garment circuit at once. The single-pin module is useful for stand-alone one-pin components like electrodes, or for components with multiple leads that are spatially separated, such as the two ends of a strip-form strain sensor. The two-pin module allows components like vibrating motors or 1-wire-bus components to be attached to the circuit with one connector: the two conductive LumeoLoop modules are separated such that the connector can be offset on the textile layup so that one module makes contact with the addressable trace and the other makes contact with the ground plane. The two non-conductive modules serve to stabilize the component in our case.

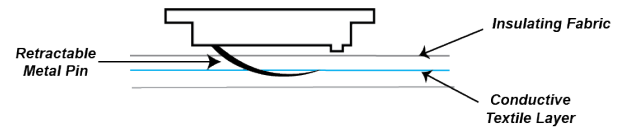


Figure 3: LumeoLoop connector mechanism

Using this e-textile architecture, we designed a lower-limb garment for rehabilitation applications (Figure 4). The garment allows components to be attached on the dorsal and ventral thighs and calves, both on the inside and outside of the garment. It is designed to support an array of rehabilitation technologies, including sensors (simple 2-lead sensors like piezoresistive strain or pressure sensors, or 1-wire bus ICs) for goniometry [9] or monitoring of swelling [11]; vibrotactile actuators for haptic cueing [12]; shape-memory actuators for active muscle support [13], management of circulation [14], or counteracting swelling; and electrodes for EMG measurement [6] or functional electrical stimulation [15]. A very wide variety of rehabilitation technologies are enabled by these sensing and actuating devices.

Addressable leads and the shared ground plane are routed to a central location on the wearer's hip, using insulated conductive thread (SewIY). They can then be attached to a variety of connectors, or directly to a PCB for integration into a system.

In prior work [16], our team conducted an electrical performance analysis in order to identify candidate materials and quantify their effects on component performance. We found that while conductive rubber worked well electrically, its mechanical properties were inadequate for maintaining the comfort and wearability of a stretch garment. Conductive silver-coated Nylon jersey (LessEMF), silver-coated Nylon thread (Sheildex), and silver-coated Vectran thread

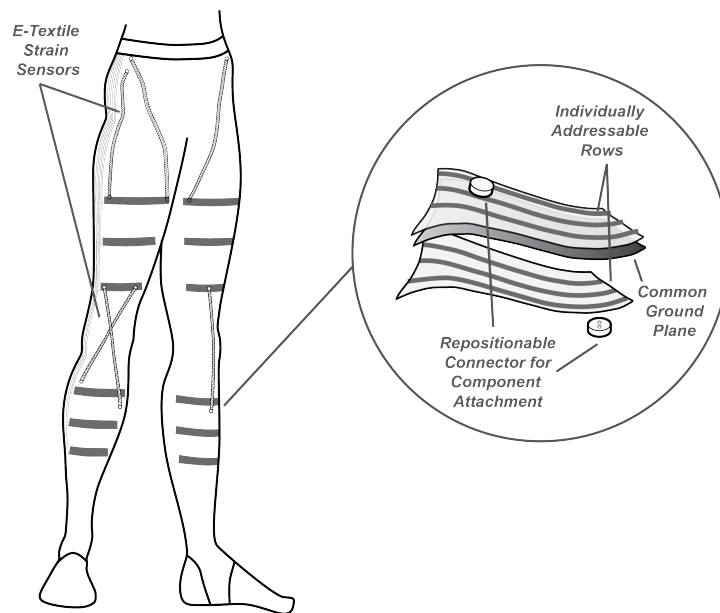


Figure 4: Lower-limb rehabilitation system design

(Syscom) all performed well electrically and had appropriate mechanical properties.

The design process for implementing this garment concept focused on refining the textile layup and manufacturing methods, with a particular focus on the addressable traces. Methods were timed and costed, and electrical performance was characterized (Table 1) for each construction method.

2.1 Stitched Leads

Two conductive threads were implemented using stitched methods to form the addressable traces: a silver-coated Nylon thread and a silver-coated Vectran thread. Addressable traces were stitched using a Brother BAS342G Pattern Stitching machine, with the conductive thread in the bobbin and polyester sewing thread in the needle. A dense serpentine pattern 3/8" wide was implemented to create a solid conductive fill for the LumeoLoop connector to make contact with (Figure 5). Traces were stitched to the polyester knit face fabric with a soluble stabilizer to support the stitching process, which was then dissolved following sewing.



Figure 5: Connection between LumeoLoop connector barb and stitched (L) and jersey (R) traces.

As seen in Table 1, the stitched construction methods were time-consuming and expensive. Further, machine faults (always a challenge when using non-traditional threads) resulted in frequent re-work and lost materials – additional expenses that are not reflected in Table 1

2.2 Laminated Leads

We also explored using the silver-coated Nylon jersey fabric used as the ground plane for the addressable traces. Rectangular pieces 3/8" wide were laser-cut, then laminated to the polyester face fabric using a fusible web adhesive. Bounding boxes for trace placement were pre-stitched on the polyester face fabric using polyester sewing thread. Prior to lamination, a square of zig zag stitching using silver-coated Vectran thread in the bobbin and polyester thread in the needle was stitched overlapping the end of the bounding area for the silver jersey trace. This is necessary so that the insulated thread used for the stitched interconnect traces can be soldered to the Vectran thread (Nylon-based conductive textiles are not temperature-stable enough to be soldered).

3 TESTING

Fabrication time and materials cost were computed for the three different fabrication methods, as shown in Table 1.

3.1 Fabrication Comparison

Costs were computed for fabrication of the full leggings prototype (as illustrated in Figure 6) containing 96 traces. The stitched Vectran approach was much more expensive than the two Nylon-based approaches, mainly driven by the cost of the Vectran thread itself (\$0.66USD/yd vs. \$0.12USD for Nylon). The silver-coated Nylon thread vs. jersey approaches were closer in cost, but the jersey approach reduced production time by about 20% (again, this also

Table 1: Characterization and performance test results

	Stitched Nylon Traces	Stitched Vectran Traces	Conductive Jersey Traces
Maximum current (Amp) at 70C (or failure)	229 (Failed)	654.66	649
Resistance Variability	29.7121	1.3964	3.2532
Cost (\$)	\$199.42	\$370.53	\$175.52
Time to manufacture traces (hh:mm:ss)	5:02:36	4:49:12	4:05:24
Connector Reliability (%)	65%	56%	81%
Connector Failure (%)	33%	17%	0%
Connection Resistance (average, ohms)	11.5	3.6	7.7

does not include machine faults, which would further increase the production time for stitched methods).

3.2 Functional Performance

Electrical and mechanical performance was assessed for each method. Variability in baseline resistance was calculated in the thermal testing by dividing the power supply voltage by the current. Stitched Vectran traces were the least variable, at approximately 1 Ohm, and Nylon jersey traces were close behind (~3 Ohms). Stitched Nylon traces had considerably more variability in resistance (~30 Ohms). Resistance variability in traces affects system performance in different ways for different components. It can be significant for components like piezoresistive sensors, where introduction of a new resistor into the circuit can change calibration of sensors.

To assess the current load each method could sustain, traces were loaded with current until they reached 70 deg C (threshold for discomfort for the wearer) or failure, whichever came first. The stitched Vectran and laminated Nylon jersey performed similarly, sustaining ~650mA of current at 70C. The stitched Nylon traces failed before reaching 70C, at 229mA. Current limits are most important for higher-current components like SMA actuators or heating pads. Based on our prior work, the better-performing traces here can support about 4N of force from an SMA spring actuator [17], or 30-40C temperature from a stitched thermal actuator [18].

To test the performance of the connectors, a repeatability test with four single-pin module connectors connected to the three different types of stitched traces was done. The connectors were attached and detached to the traces repeatedly (9 times per type of trace, as shown in Figure 5) and the number of times the connectors made a successful electrical connection was recorded. As seen in Table 1, the conductive jersey performed the best with the connectors making contact 81% of the time. The vectran and the stitched nylon traces were similar in terms of connector reliability percentage, however, the module failure rate (33%) was higher in nylon traces than the vectran.

4 FINAL SYSTEM IMPLEMENTATION

The final leggings prototype was implemented using laminated conductive jersey traces, as shown in Figure 6. Figure 7 shows a stitched strain sensor [10] attached in multiple configurations, using just the conductive connectors (Figure 7 right) and using conductive connectors at the ends with non-conductive connectors in the middle to control sensor contour over the knee surface (Figure

**Figure 6: Final garment system implementation**

7 left). Figure 8 shows the insulated lead routing to the wearer's hip, as well as detail of the external appearance of the addressable traces and single-pin connector attachment. As previously mentioned, no conductive components are exposed on the outside of the layup. Non-conductive stitching shows the location of the addressable traces. Ground plane connections can be made between the addressable traces.



Figure 7: Flexible sensor positioning using conductive connectors(R) and both conductive and non-conductive connectors (L)



Figure 8: Trace routing (L) and strain sensor attachment to ground plane (R)

5 CONCLUSION

Flexibility in form and function of wearable systems enables easier fitting and efficient adaptation of functionality. To date, efforts in this vein have focused on hardware building kits, which do not enable easy wearability. Here, we present the first system that enables both custom placement of components and adaptation of functionality on the body surface. The resulting garment preserves the appearance and usability of leggings, while enabling flexibility of placement and functionality.

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